Unit 4 Progress Check: MCQ Part B

1. In the following code segment, a valid Scanner object named scan is created to read from the text file input.txt.

```
File readFile = new File("input.txt");
Scanner scan = new Scanner(readFile);

int firstIn = scan.nextInt();
int secondIn = scan.nextInt();
int thirdIn = scan.nextInt();
```

Which of the following could be the contents of input.txt so that the variables firstIn, secondIn, and thirdIn are each assigned a valid value by the code segment?

(A)

```
123
```

(B)

```
1
2
3
```

(C)

```
1, 2, 3
```

(D)

```
firstIn: 1 secondIn: 2 thirdIn: 3
```

2. In the following code segment, a valid `Scanner` object named `fileInput` is created to read from the text file `numbers.txt`. Assume that the text file contains only numerical data and spaces.

```
File numData = new File("numbers.txt");
Scanner fileInput = new Scanner(numData);

double sum = 0.0;
while (fileInput.hasNext())
{
  /* missing code */
}

System.out.println(sum);
```

Which of the following can be used to replace `/* missing code */` so that the code segment prints the sum of the positive numbers in the file?

Unit 4 Progress Check: MCQ Part B

(A)

```
if (fileInput > 0)
{
  sum = sum + fileInput;
}
```

(B)

```
if (fileInput.nextDouble() > 0)
{
  sum = sum + fileInput.nextDouble();
}
```

(C)

```
double temp = fileInput.nextDouble();
if (temp > 0)
{
  sum = sum + temp;
}
```

(D)

```
double temp = fileInput.nextDouble();
if (temp > 0)
{
  sum = sum + fileInput.nextDouble();
}
```

3. A text file named data.txt has the following contents. Assume that the file contains no spaces.

```
123
456
789
```

In the following code segment, a valid Scanner object named source is created to read from the text file.

```
int x = 0;

File text = new File("data.txt");
Scanner source = new Scanner(text);
```

Unit 4 Progress Check: MCQ Part B

```
while (source.hasNext())
{
  x = source.nextInt();
}
System.out.println(x);
```

What is printed as a result of executing the code segment?

(A)
0

(B)
123

(C)
789

(D)
123456789

4. Which of the following statements will store the value 34.0 in the double variable value?

(A)
```
double value = (double) "34";
```

(B)
```
double value = Double.parseDouble("34");
```

(C)
```
double value = Integer.parseInt("34") + ".0";
```

(D)
```
double value = Double.parseDouble("34") + ".0";
```

5. Consider the following calculate method.

```
public double calculate(double x)
{
  return x + 1.5;
}
```

The following code segment calls the method calculate in the same class.

Unit 4 Progress Check: MCQ Part B

```
Double d1 = new Double(7.5);
System.out.println(calculate(d1));
```

What, if anything, is printed when the code segment is executed?

(A)
8.5

(B)
9

(C)
9.0

(D) Nothing is printed because the code does not compile. The actual parameter d1 passed to calculate is a Double, but the formal parameter x is a double.

6. A text file named grades.txt has the following contents.

```
89 92 90 88 79 85 77
```

In the following code segment, a valid Scanner object named scan is created to read from the text file. The code segment is intended to store the values from grades.txt into an int array.

```
File gradeFile = new File("grades.txt");
Scanner scan = new Scanner(gradeFile);

int[] reportCard = new int[7];
for (int i = 0; i < reportCard.length; i++)
{
  int currentGrade = /* missing expression */ ;
  reportCard[i] = currentGrade;
}
```

Which of the following can replace /* missing expression */ so that the code segment works as intended?

(A)
Double.parseDouble(scan.next())

(B)
Integer.parseInt(scan.next())

(C)
Integer.parseInt(scan.nextInt())

Unit 4 Progress Check: MCQ Part B

(D)

```
Integer.parseInt(scan.nextLine())
```

7. Consider the following statement, which is intended to create an ArrayList named al to store only elements of type Thing. Assume that the Thing class has been properly defined and includes a no-parameter constructor.

```
ArrayList<Thing> al = /* missing code */
```

Which of the following can be used to replace /* missing code */ so that the statement works as intended?

(A)

```
new Thing();
```

(B)

```
new ArrayList<Thing>();
```

(C)

```
new ArrayList(Thing);
```

(D)

```
new ArrayList<>(Thing);
```

8. Consider the following code segment.

```
ArrayList<Integer> ages = new ArrayList<Integer>();
ages.add(18);
ages.add(20);
ages.add(1, 15);
```

What are the contents of ages after executing the code segment?

(A)

[15, 18, 20]

(B)

[18, 15, 20]

(C)

[18, 20, 15]

(D)

[20, 15, 18]

Unit 4 Progress Check: MCQ Part B

9. Consider the following code segment.

```
ArrayList<Integer> nums = new ArrayList<Integer>();

nums.add(3);
nums.add(2);
nums.add(1);
nums.add(0);
nums.add(0, 4);
nums.set(3, 2);
nums.remove(3);
nums.add(2, 0);
```

Which of the following represents the contents of nums after the code segment is executed?

(A)
[2, 3, 2, 1, 0]

(B)
[4, 0, 0, 1, 3]

(C)
[4, 2, 0, 2, 0]

(D)
[4, 3, 0, 2, 0]

10. Consider the following method.

```
public static void processNumbers(ArrayList<Integer> numbers)
{
   int last = numbers.size() - 1;
   numbers.add(0, numbers.get(last));
   numbers.remove(last + 1);
}
```

Which of the following best describes the preconditions necessary for the method to run without error?

(A) The `ArrayList` must be sorted from smallest to largest.
(B) The `ArrayList` must contain at least one element.
(C) The `ArrayList` must contain at least two elements.
(D) The `ArrayList` must contain only positive `Integer` values.

Unit 4 Progress Check: MCQ Part B

11. Consider the following code segment.

```
ArrayList<Integer> nums = new ArrayList<Integer>();

nums.add(12);
nums.add(13);
nums.add(15);
nums.add(10);

if (nums.get(0) > nums.get(nums.size() - 1))
{
  int x = nums.get(0);
  nums.set(0, nums.get(nums.size() - 1));
  nums.set(nums.size() - 1, x);
}
```

Which of the following best describes the behavior of the code segment?

(A) It compares all values in nums and places nums in order from smallest to largest.
(B) It compares all values in nums and places nums in order from largest to smallest.
(C) It compares the first and last elements in nums and swaps them if the first element is larger than the last.
(D) It compares the first and last elements in nums and swaps them if the first element is smaller than the last.

12. In the following code segment, `numbers` is a properly declared `ArrayList` variable that has been initialized with the following values.

```
[13, 2, -4, 0, 34, -12, -13, 18, -27]
```

```
for (int i = 0; i < numbers.size(); i++) // Line 1
{
  if (numbers.get(i) < 0)
  {
    numbers.remove(i);
  }                    // Line 6
  i--;                 // Line 7
}
```

The code segment is intended to remove all negative values from `numbers`. However, the code segment is not working as intended.

Which of the following changes can be made so that the code segment works as intended?

(A) In line 1, changing `i < numbers.size()` to `i < numbers.size() - 1`
(B) In line 7, changing `i--;` to `i++;`
(C) Removing line 7
(D) Switching lines 6 and 7

Unit 4 Progress Check: MCQ Part B

13. In the following code segment, intList is properly declared and initialized as an ArrayList<Integer> variable.

```
int someVal = 0;

for (int i = 0; i < intList.size(); i += 2)
{
  someVal++;
}

System.out.println(someVal);
```

Which of the following best describes the value printed as a result of executing the code segment?

(A) The sum of all the elements in intList
(B) The number of elements in intList
(C) The number of even values in intList
(D) The number of elements in even-numbered indexes in intList

14. Consider the following code segment.

```
ArrayList<Integer> myList = new ArrayList<Integer>();
myList.add(75);
myList.add(20);
myList.add(99);

ArrayList<Integer> yourList = new ArrayList<Integer>();
/* missing code */
```

After the code segment is executed, yourList should contain [20, 99, 75].

Which of the following can replace /* missing code */ to produce this result?

(A)

```
int initialSize = myList.size();
for (int i = 0; i < initialSize; i++)
{
  myList.add(2, myList.remove(0));
  yourList.add(myList.get(0));
}
```

Unit 4 Progress Check: MCQ Part B

(B)

```
int initialSize = myList.size();
for (int i = 1; i <= initialSize; i++)
{
  yourList.add(myList.get(i));
}
```

(C)

```
for (Integer temp : myList)
{
  yourList.add(0, temp);
}
```

(D)

```
for (Integer temp : myList)
{
  myList.add(2, myList.remove(0));
  yourList.add(temp);
}
```

15. In the following code segment, numbers is a properly declared and initialized ArrayList<Integer> variable.

```
for (int i = 0; i < numbers.size(); i++)
{
  if (numbers.get(i) < 0)
  {
    numbers.remove(i);
  }
}
```

The code segment is intended to remove all negative values from numbers. Which of the following best describes the conditions in which the code segment works as intended?

(A) The code segment works as intended for all possible contents of numbers.
(B) The code segment works as intended only when negative numbers are not in adjacent positions in numbers.
(C) The code segment works as intended only when numbers is sorted.
(D) The code segment works as intended only when there are no duplicate values in numbers.

16. Consider the following method.

Unit 4 Progress Check: MCQ Part B

```
public void removeAll(int target, ArrayList<Integer> list)
{
  for (int index = list.size(); index >= 0; index--)  // Line 3
  {
    if (list.get(index) == target)                    // Line 5
    {
      list.remove(index);
    }
  }
}
```

The method is intended to remove all values matching a target value from its `ArrayList` parameter. However, it is not working as intended.

Which of the following changes can be made so that the method works as intended?

(A) In line 3, changing the loop initialization to `index = list.size() - 1`
(B) In line 3, changing the loop condition to `index > 0`
(C) In line 3, changing the loop increment to `index++`
(D) In line 5, changing the list access expression to `list.get(target)`

17. The following code segment initializes and processes an `ArrayList` of `String` elements. The loop is intended to delete every other element in the `ArrayList` starting with the first element.

```
ArrayList<String> nameList = new ArrayList<String>();
nameList.add("Stacey");
nameList.add("Bob");
nameList.add("Susan");
nameList.add("Pete");

for (int i = 0; i < nameList.size(); i++)
{
  /* missing code */
}
```

After the code segment is executed, `nameList` should contain `["Bob", "Pete"]`.

Which of the following can replace `/* missing code */` so that the code segment works as intended?

(A)
```
nameList.remove(i);
```

(B)
```
nameList.remove(i);
i++;
```

Unit 4 Progress Check: MCQ Part B

(C)

```
nameList.set(i, "Bob");
nameList.set(i, "Pete");
```

(D)

```
nameList.set(i, "Bob");
nameList.set(i + 1, "Pete");
```

18. Consider the following method.

```
public static void transform(ArrayList<Integer> items)
{
  for (int i = items.size() - 1; i >= 0; i--)
  {
    items.add(i, items.get(i));
  }
}
```

The following code segment appears in a method in the same class as `transform`.

```
ArrayList<Integer> numbers = new ArrayList<Integer>();
numbers.add(15);
numbers.add(4);
numbers.add(19);
numbers.add(7);

transform(numbers);
```

What are the contents of `numbers` after executing the code segment?

(A)

[15, 4, 4, 19, 19, 7, 7]

(B)

[15, 4, 19, 7, 7, 19, 4, 15]

(C)

[15, 4, 19, 7, 15, 4, 19, 7]

(D)

[15, 15, 4, 4, 19, 19, 7, 7]

19. Consider the following code segment.

Unit 4 Progress Check: MCQ Part B

```
double x = 0.0;
for (double number : numbers)
{
   x += number;
}
System.out.println(x / numbers.size());
```

Assume that numbers is an ArrayList that has been initialized with the following Double objects.

[10.0, 20.0, 30.0, 40.0, 100.0]

What is printed as a result of executing the code segment?

(A)
20.0

(B)
30.0

(C)
40.0

(D)
200.0

20. In the following code segment, words is an ArrayList of two or more String objects.

```
int prev = words.get(0).length();
int count = 0;

for (int j = 1; j < words.size(); j++)
{
  int curr = words.get(j).length();
  if (curr == prev)
  {
    count += 1;
  }
  prev = curr;
}
```

Which of the following best describes the value assigned to count as a result of executing the code segment?

(A) The number of elements in words whose length is the same length as the first element
(B) The number of elements in words whose length is the same length as the element immediately preceding it
(C) The number of elements in words whose value is equal to the first element

Unit 4 Progress Check: MCQ Part B

(D) The number of elements in words whose value is equal to the element immediately preceding it

21. In the code segment shown, `myList` is an `ArrayList` of integers. The code segment is intended to remove all elements with the value `0` from `myList`.

```
int j = 0;
while (j < myList.size())
{
  if (myList.get(j) == 0)
  {
    myList.remove(j);
  }
  j++;
}
```

The code segment does not always work as intended.

For which of the following lists does the code segment **not** produce the correct result?

(A)
[0, 1, 0, 2]

(B)
[1, 0, 0, 2]

(C)
[1, 2, 3, 0]

(D)
[1, 2, 3, 4]